

~~A~~ Part A - Heuristic Evaluation

1. Problem Statement

The chosen website for evaluation is Prothom alo (<https://www.prothomalo.com>), one of Bangladesh leading online news portals. It provides national and international news, opinion pieces, entertainment and multimedia content in Bangla. The website is widely used but has certain usability challenges such as dense layouts, unclear navigation, and inconsistent visual hierarchy.

Evaluator 1 - Md. Ahsan Kabir

Usability Problem Description	Violated Heuristic Principle (s)	Rating
☐ Homepage Overcrowded with text and ads making it difficult to locate key section	Aesthetic and Minimalist Design	4

Navigation bar items not
clearly categorized user
has to scroll to find
topics.

Match Between System 3
Real world

font size on article
pages too small for
long reading sessions.

visibility of system 4

Inconsistent placement of
images and text on
mobile view

consistency and 3
standards

Advertisement pop-ups
interfere with content
reading experience

Error Prevention 5

Evaluator 2: Efat Jahan Ema

Usability Problem Description	Violated Heuristic Principle	Rating
Search function often returns unrelated result or duplicates	Error Prevention	4
Hyperlinks are not visually distinct from regular text	Recognition Rather than recall	3
No clear visual hierarchy on home page, headings blend together	Aesthetic and Minimalist Design	4
Slow page load due to excessive ads scripts	Efficiency of Use	5
No clear feedback when a link is clicked or loading	Visibility of System Status	3

Final consolidated Table

Rank	Usability Problem	Violated Principle	Severity
1.	Advertisement Pop-ups disrupt reading flow	User control & freedom	5
2.	Slow page load and ad-heavy layout	Efficiency of use	5
3.	Small article font size difficult readability	Aesthetic and Minimalist Design	4
4.	Unclear navigation and crowded homepage	Match Between System and Real world	4
5.	Weak visual hierarchy among headlines	Consistency Standards	3

After discussion both evaluators ~~aggr~~ agreed that ad interruptions and slow performance are the most critical issues impacting usability. Problems were prioritized based on frequency user impact and fix complexity. Readability issues ranked next due to their direct effect on content engagement.

Part - B

1. Problem Statement

The A/B test focuses on improving the readability of sample news article page. Currently articles have small text dense paragraphs and little white space, which can strain readers eyes.

2. Chosen Feature

Feature: Article layout and typography.

This is ideal for A/B testing because

small design changes like font size line height and margin spacing can be measured using metrics such as average time on page and scroll depth.

3. Two Alternative Designs

Design A - Current article layout (small font dense text)

Design B - Improved layout (larger font increased spacing, simplified sidebars)

4. Test Description

Users would be randomly assigned to

either Design A or Design B. Engagement

metrics such as average time spent reading scroll depth and bounce rate would be recorded to measure improvement.

5. Expected Results and Hypothesis

Hypothesis: Design B will improve readability and engagement. We expect a 15-20% increase in average time on page and reduced bounce rate. Readers are likely to find the cleaner design more comfortable for extended reading.
